

October 21, 2022

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: LLHg
Pace Project No.: 50328165

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on October 12, 2022. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,



Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: LLHg
Pace Project No.: 50328165

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268
Illinois Accreditation #: 200074
Indiana Drinking Water Laboratory #: C-49-06
Kansas/TNI Certification #: E-10177
Kentucky UST Agency Interest #: 80226
Kentucky WW Laboratory ID #: 98019

Michigan Drinking Water Laboratory #9050
Ohio VAP Certified Laboratory #: CL0065
Oklahoma Laboratory #: 9204
Texas Certification #: T104704355
Wisconsin Laboratory #: 999788130
USDA Soil Permit #: P330-19-00257

REPORT OF LABORATORY ANALYSIS

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SAMPLE SUMMARY

Project: LLHg
Pace Project No.: 50328165

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50328165001	Field Blank	Water	10/11/22 13:35	10/12/22 10:18
50328165002	Effluent	Water	10/11/22 13:40	10/12/22 10:18
50328165003	Effluent Duplicate	Water	10/11/22 13:43	10/12/22 10:18
50328165004	Influent	Water	10/11/22 07:00	10/12/22 10:18

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SAMPLE ANALYTE COUNT

Project: LLHg
Pace Project No.: 50328165

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50328165001	Field Blank	EPA 1631E	WJW	1	PASI-I
50328165002	Effluent	EPA 1631E	WJW	1	PASI-I
50328165003	Effluent Duplicate	EPA 1631E	WJW	1	PASI-I
50328165004	Influent	EPA 245.1	EAE	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

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SUMMARY OF DETECTION

Project: LLHg
Pace Project No.: 50328165

Lab Sample ID Method	Client Sample ID Parameters	Result	Units	Report Limit	Analyzed	Qualifiers
50328165002	Effluent					
EPA 1631E	Mercury	1.02	ng/L	0.50	10/20/22 09:38	
50328165003	Effluent Duplicate					
EPA 1631E	Mercury	0.860	ng/L	0.50	10/20/22 09:31	
50328165004	Influent					
EPA 245.1	Mercury	0.39	ug/L	0.20	10/19/22 16:42	

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ANALYTICAL RESULTS

Project: LLHg
Pace Project No.: 50328165

Sample: Field Blank		Lab ID: 50328165001		Collected: 10/11/22 13:35		Received: 10/12/22 10:18		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
1631E Mercury, Low Level		Analytical Method: EPA 1631E Preparation Method: EPA 1631E Pace Analytical Services - Indianapolis							
Mercury	<0.25	ng/L	0.50	0.25	1	10/19/22 11:43	10/20/22 07:32	7439-97-6	

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ANALYTICAL RESULTS

Project: LLHg
Pace Project No.: 50328165

Sample: Effluent		Lab ID: 50328165002		Collected: 10/11/22 13:40		Received: 10/12/22 10:18		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
1631E Mercury, Low Level		Analytical Method: EPA 1631E Preparation Method: EPA 1631E Pace Analytical Services - Indianapolis							
Mercury	1.02	ng/L	0.50	0.25	1	10/19/22 11:43	10/20/22 09:38	7439-97-6	

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ANALYTICAL RESULTS

Project: LLHg
Pace Project No.: 50328165

Sample: Effluent Duplicate		Lab ID: 50328165003	Collected: 10/11/22 13:43	Received: 10/12/22 10:18	Matrix: Water				
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
1631E Mercury, Low Level		Analytical Method: EPA 1631E Preparation Method: EPA 1631E Pace Analytical Services - Indianapolis							
Mercury	0.860	ng/L	0.50	0.25	1	10/19/22 11:43	10/20/22 09:31	7439-97-6	

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ANALYTICAL RESULTS

Project: LLHg
Pace Project No.: 50328165

Sample: Influent		Lab ID: 50328165004		Collected: 10/11/22 07:00		Received: 10/12/22 10:18		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
245.1 Mercury Analytical Method: EPA 245.1 Preparation Method: EPA 245.1 Pace Analytical Services - Indianapolis									
Mercury	0.39	ug/L	0.20	0.091	1	10/19/22 10:24	10/19/22 16:42	7439-97-6	

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QUALITY CONTROL DATA

Project: LLHg
Pace Project No.: 50328165

QC Batch:	701713	Analysis Method:	EPA 1631E
QC Batch Method:	EPA 1631E	Analysis Description:	1631E Mercury
		Laboratory:	Pace Analytical Services - Indianapolis
Associated Lab Samples: 50328165001, 50328165002, 50328165003			

METHOD BLANK: 3225739 Matrix: Water
Associated Lab Samples: 50328165001, 50328165002, 50328165003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	ng/L	<0.25	0.50	0.25	10/20/22 07:15	

METHOD BLANK: 3225740 Matrix: Water
Associated Lab Samples: 50328165001, 50328165002, 50328165003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	ng/L	<0.25	0.50	0.25	10/20/22 08:11	

METHOD BLANK: 3225741 Matrix: Water
Associated Lab Samples: 50328165001, 50328165002, 50328165003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	ng/L	<0.25	0.50	0.25	10/20/22 10:54	

LABORATORY CONTROL SAMPLE: 3225742

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	ng/L	5	4.81	96	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3225743 3225744

Parameter	Units	50328125002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	ng/L	<0.50	2.5	2.5	2.34	2.40	94	96	71-125	3	24	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3225923 3225924

Parameter	Units	50328228002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	ng/L	1.00	4	4	4.77	4.80	94	95	71-125	1	24	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

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QUALITY CONTROL DATA

Project: LLHg
Pace Project No.: 50328165

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3226205 3226206												
Parameter	Units	50328109020 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	ng/L	1.03	4	4	4.69	5.02	92	100	71-125	7	24	

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QUALITY CONTROL DATA

Project: LLHg
Pace Project No.: 50328165

QC Batch:	701443	Analysis Method:	EPA 245.1
QC Batch Method:	EPA 245.1	Analysis Description:	245.1 Mercury
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50328165004

METHOD BLANK: 3224679 Matrix: Water

Associated Lab Samples: 50328165004

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	ug/L	<0.091	0.20	0.091	10/19/22 15:53	

LABORATORY CONTROL SAMPLE: 3224680

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	ug/L	5	5.2	103	85-115	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3224681 3224682

Parameter	Units	50328153003 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	ug/L	0.77	5	5	5.7	5.4	98	94	70-130	4	20	

MATRIX SPIKE SAMPLE: 3224683

Parameter	Units	50328197002 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	ug/L	ND	5	5.1	102	70-130	

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QUALIFIERS

Project: LLHg
Pace Project No.: 50328165

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: LLHg
Pace Project No.: 50328165

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50328165001	Field Blank	EPA 1631E	701713	EPA 1631E	701940
50328165002	Effluent	EPA 1631E	701713	EPA 1631E	701940
50328165003	Effluent Duplicate	EPA 1631E	701713	EPA 1631E	701940
50328165004	Influent	EPA 245.1	701443	EPA 245.1	701640

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Sample Conditions Upon Receipt Form (SCUR)

Date/Time: 10/12 1310	Evaluated By: DAS	WO#: 50328165 PM: BJH Due Date: 10/26/22 CLIENT: GR-Cadillac	
Client: City of Cadillac	PM: BJH		
Lab Notified of Rush or Short Holds: YES NO			
Project Received Via: FedEx UPS Client Pace Courier Other: _____		Comments:	
Custody Seal Present and Intact:	YES NO NA		
Received Sample Information Form (SIF): Drinking Waters Only	YES NO NA		
Short Hold Present (≤ 48 Hours):	YES NO		
Sample Received in Hold:	YES NO		
Custody Signature Present:	YES NO		
Collector Signature Present:	YES NO		
Sample Collected Today and On Ice:	YES NO N/A		
IR Gun #: 280 281	Temp. should be 0°C - 6°C (Initial/Corrected)		
Ice Type: WET Bagged / WET Loose BLUE NONE	1. Cooler Temp. Upon Receipt: 20.5 / 20.3		
Ice Location: TOP BOTTOM MIDDLE DISPERSED	2. Cooler Temp. Upon Receipt: _____ °C		
Temp Blank Received:	YES NO		
Sample Label Matches COC (ID/Date/Time):	YES NO		
Container Intact:	YES NO		
Correct Container:	YES NO		
Sufficient Volume:	YES NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation pH Strip Lot #: HC291590 Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl	YES NO N/A		
Residual Chlorine Absent: Cl ₂ Strip Lot #: _____ Applies to SVOC 625, PCB/Pest. 608, Total/Amenable Cyanide	YES NO N/A		
VOA Headspace Acceptable (<6mm):	YES NO N/A		
Trip Blank Received: HCl MeOH Other: _____	YES NO ON HOLD		
Comments:	3. Cooler Temp. Upon Receipt: _____ °C		
	4. Cooler Temp. Upon Receipt: _____ °C		
	Non-Conformance Form Required: YES NO		